Zeitschr. f. math. Logik und Grundlagen d. Math.
Bd. 7, S. 124—135 (1961)

ON THE CONSISTENCY AND UNDECIDABILITY OF RECURSIVE ARITHMETIC

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1. Introduction

In this paper, we shall give a proof of the consistency of primitive recursive arithmetic (hereafter referred to as R) and deduce GÖDELS two incompleteness results for R. We shall show that, by taking a particular finite codification of R, using function variables, the proof of freedom from contradiction for systems similar to R, given by HILBERT and BERNAYS [3, see bibliography], can be carried out in a meta-language R' which is itself a form of recursive arithmetic. (R' contains R and one function defined by a double recursion, which effectively gives an enumeration of the functions of R.) Following this we shall show that the proposition 'R is free from contradiction' is undecidable in R.

The manner of proof of the results in this paper establishes a technique which may be used to derive similar results for a class of recursive arithmetics. If R_a and R_b are recursive arithmetics with the following properties:

(a) R_a contains R,

(b) R_a may be given a finite codification by introducing function variables to the primitive basis of the system; that is, by introducing variables for functions of one argument place, variables for functions of two argument places, . . ., variables for functions of n argument places, for some finite number n,

(c) the axioms of R_a are finite in number and, upon arbitrary substitution of numerals for numerical variables and functions for function variables, are verifiable,

(d) the axiom schemas of R_a are those of R and rules for substitution for the function variables,

(e) R_b contains R_a and the functions Φ_i $(i = 1, 2, \ldots, n)$, where Φ_i gives an enumeration of the i-th argument place functions of R_a,

and C stands for the proposition expressing freedom from contradiction for R_a, then,

(1) C belongs to R_a,
(2) C is provable in R_b,
(3) C is undecidable in R_a

In particular, it follows that the freedom from contradiction of k-recursive arithmetic (in the sense of R. PÉTER [5]) is provable in $(k+1)$-recursive arithmetic. In a paper to appear shortly we shall show how these results may be used to establish the independence of certain induction schemas in recursive arithmetic.

During the preparation of this paper I have drawn freely from a manuscript by Professor ALONZO CHURCH, in which a sketch of the proof of the freedom from contradiction of his codification of primitive recursive arithmetic (known as Binary Recursive Arithmetic, see [1]) is given; this result is the one he discussed at the Edinburgh International Congress of Mathematicians 1958 (see [8, p. 6]). I wish to record my sincere thanks to Professor CHURCH for allowing me access to his work and to Professor R. L. GOODSTEIN for much help and guidance.

2. Description of Ternary Recursive Arithmetic

Ternary Recursive Arithmetic is a codification of primitive recursive arithmetic as a logic-free equation calculus with function variables, in which there are two basic entities, functions and terms. The functions have one, two or three argument places and are defined by recursion or composition and the terms are defined explicitly, from the functions, but have no function sign attached to them. By this device primitive recursive arithmetic without function variables is given a finite codification. The version of this system given in [6] has been slightly improved, mainly by changes in axiom schema U and axiom 10.

The primitive symbols of the system are (a) infinite lists of numerical variables $a, b, c, \ldots$, 1-function variables $f, g, h, \ldots$, 2-function variables $f', g', h', \ldots$, and 3-function variables $f'', g'', h'', \ldots$; (b) the constant functors s, o, i, j and k; (c) the functor connectives R, R', C, C' and K; (d) the numerical constant 0; (e) the connective $=$; (f) brackets, commas and parentheses. The functors are defined by the following scheme,

(i) A *1-functor* is one of the following: — the constant functors s, o or i; a 1-function variable standing alone; Kf where f is a 2-functor.

(ii) A *2-functor* is one of the following: — the constant functor j; a 2-function variable standing alone; $R'fg$ where f is a 1-functor and g is a 2-functor; $C'fgh$ where f is a 3-functor, g is a 2-functor and h is a 1-functor.

(iii) A *3-functor* is one of the following: — the constant functor k; a 3-function variable standing alone; $Rfgh$ where f and g are 2-functors and h is a 3-functor; $Cpfgh$ where p, f, g and h are 3-functors.

A *1-function* is a symbol of the form $f(a)$ where f is a 1-functor and a is a numerical variable, similarily a *2-function* has the form $f(a, b)$ and a *3-function* has the form $f(a, b, c)$. A *term* is one of the following: — (a) the numerical constant 0; (b) a numerical variable standing alone; (c) $f(A)$ where f is a 1-functor and A is a term; (d) $f(A, B)$ where f is a 2-functor and A and B are terms; (e) $f(A, B, C)$ where f is a 3-functor and A, B and C are terms. The expression $A = B$, where A and B are terms, is called an *equation*.

The axiom schema of the system are labelled E, F, T, U, V and W and are as follows,

E. $f(sa, b, c) = f(a, b, c) \vdash f(a, b, c) = f(0, b, c)$, where f is a 3-functor.

F. $f(0, b) = g(0, b), f(sa, b) = C'pfo(a, b), g(sa, b) = C'pgo(a, b) \vdash f(a, b) = g(a, b)$, where f and g are 2-functors, p is a 3-functor and o is defined by axiom 1.

T. $A = B$, $A = C \vdash B = C$, where A, B and C are terms.

U. $A = B \vdash f(A, b, c) = f(B, b, c)$, where A and B are terms, b and c are variables and f is a 3-functor.

V. If an equation involving a numerical variable $\boldsymbol{a}$ is represented by $P(\boldsymbol{a})$ and the result of replacing all occurrences of $\boldsymbol{a}$ by A (a term) in $P(\boldsymbol{a})$ is denoted by $P(A)$, then $P(\boldsymbol{a}) \vdash P(A)$. ($\boldsymbol{a}$ is a meta-variable.)

W. If an equation involving a function variable $\boldsymbol{f}$ is represented by $P\boldsymbol{f}$ and the result of replacing all occurrences of $\boldsymbol{f}$ by g (a functor) in $P\boldsymbol{f}$ is denoted by $P\boldsymbol{g}$, then $P\boldsymbol{f} \vdash P\boldsymbol{g}$ where $\boldsymbol{f}$ is a function variable and $\boldsymbol{g}$ is a functor belonging to the same class as $\boldsymbol{f}$. ($\boldsymbol{f}$ is a meta-variable.)

Finally the axioms are given as follows,

1. $o(a) = 0$
2. $i(a) = a$
3. $j(a, b) = b$
4. $k(a, b, c) = c$
5. $Rfgh(0, b, c) = f(b, c)$
6. $Rfgh(sa, b, c) = g(h(a, b, c), Rfgh(a, b, c))$
7. $Cpfgh(a, b, c) = p(f(a, b, c), g(a, b, c), h(a, b, c))$
8. $R'fg(0, b) = f(b)$
9. $R'fg(sa, b) = g(a, R'fg(a, b))$
10. $C'fgh(a, b) = f(g(a, b), b, h(a))$
11. $Kf(a) = f(a, 0)$.

The derivation of primitive recursive arithmetic from this codification is described in [6]. The main point in this derivation, once addition, modified subtraction and multiplication are defined and their basic properties are proved, is to introduce the notion of an ordered pair: that is we define the functions $w(a, b)$ [$w(a, b)$ is the triangular number of a and b], $m_1 a$ and $m_2 a$, and prove

$$m_1 w(a, b) = a, \quad m_2 w(a, b) = b, \quad w(m_1 a, m_2 a) = a.$$

From this we can show that the class of primitive recursive functions is the same as the class of terms in Ternary Recursive Arithmetic.

In a standard manner the inequality relations and the connectives of the propositional calculus ($\neg$ — not, & — and, $\rightarrow$ — implies) are introduced contexturally. The following functions are also defined,

$a(a, b) = 0$ if $a = b$, and 1 otherwise.

$q(a, b)$ is the quotient given when b is divided into a.

$r(a, b)$ is the remainder left when b is divided into a.

$\sum\limits_{t}^{x} [f(t)]$ is the sum $f(0) + f(1) + \cdots + f(x)$.

The basic properties of all these functions and relations are clear, they will be used tacitly in the following work; see [2] for a detailed analysis of them.

3. Proof of Freedom from Contradiction of Ternary Recursive Arithmetic

The proof of freedom from contradiction of Ternary Recursive Arithmetic is carried out in a system consisting of Ternary Recursive Arithmetic with one additional axiom. This additional axiom is a definition of the function $vf(x, y)$ from which the functions $vt(x, y)$, $vb(x, y)$ and $vu(x, y)$ can be defined and these give an enumeration of the functions of the original system. The function $vf(x, y)$ is doubly recursive and performs a similar rôle to the R. PÉTER example of a doubly recursive function which enumerates all the functions of primitive recursive arithmetic. For future reference we label Ternary Recursive Arithmetic 'system T' and the extended system in which we prove freedom from contradiction of T 'system T^*'.

Freedom from contradiction of system T is proved in the following manner. First the objects of T are correlated with the natural numbers and then a valuation theory for theorems of T is set up in T^*, that is a function $vl(x, y)$ is defined to be 0 or 1 according as the equation number y for the valuation number x of its numerical variables is true or false and a second function $th(y)$ is defined to be the number of the yth theorem in some enumeration of the theorems of the system. Following this

$$vl(x, th(y)) = 0$$

is proved in T^*; that is, we have a device for proving all the theorems of the system T. From this result freedom from contradiction is easily deduced (see the end of this section).

3.1 We begin the formal proof by numbering off the objects of the system T. The numbering scheme used here is, in principle, similar to that used by GÖDEL, but it has been considerably modified for Ternary Recursive Arithmetic. 1-functors, 2-functors, 3-functors, terms and equations each have a separate numbering scheme. We list the objects in the first column and their numbers in the second, using the same letter to denote both the object and its number, no ambiguity will arise as the former are members of T and the latter members of T^*

1-functors	o	0
	i	1
	s	2
	Kg	$g + 3$
2-functors	j	0
	$R'fg$	$2w(f, g) + 1$
	$C'fgh$	$2w(f, w(g, h)) + 2$
3-functors	k	0
	$Rfgh$	$2w(f, w(g, h)) + 1$
	$Cpfgh$	$2w(p, w(f, w(g, h))) + 2$
Terms	0	0
	nth variable	$4n$
	$f(A)$	$4w(f, A) + 1$
	$g(A, B)$	$4w(g, w(A, B)) + 2$
	$h(A, B, C)$	$4w(h, w(A, w(B, C))) + 3$
Equations	$A = B$	$w(A, B)$

The numerical variables are ordered as follows $a, b, c, a_1, b_1, c_1, a_2, b_2, \ldots$.

3.2 We introduce now the simultaneous recursion equations for $vu(x, y)$, $vb(x, y)$ and $vt(x, y)$. The definition of $vf(x, y)$ in standard double recursive form and the derivation of the equations below from it using the methods of primitive recursive arithmetic only (course-of-values recursion and a theorem of R. PÉTER on the normal form of double recursive functions, given in [5]) are too long to give here.

$$vu(x, 0) = 0$$
$$vu(x, 1) = x$$
$$vu(x, 2) = x + 1$$
$$vu(x, y + 3) = vb(w(x, 0), y)$$

$$vb(x, 0) = m_2 x$$
$$vb(x, 2y + 1) = vu(m_2 x, m_1 y) \qquad \text{if } m_1 x = 0$$
$$vb(x, 2y + 1) = vb(w(m_1 pr(x), vb(pr(x), 2y + 1)), m_2 y) \qquad \text{if } m_1 x > 0$$
$$vb(x, 2y + 2) = vt(w(vb(x, m_1 m_2 y), w(m_2 x, vu(m_1 x, m_2 m_2 y))), m_1 y)$$

$$vt(x, 0) = m_2 m_2 y$$
$$vt(x, 2y + 1) = vb(m_2 x, m_1 y) \qquad \text{if } m_1 x = 0$$
$$vt(x, 2y + 1) = vb(w(vt(pr(x), m_2 m_2 y), vt(pr(x), 2y + 1)), m_1 m_2 y) \qquad \text{if } m_1 x > 0$$
$$vt(x, 2y + 2) = vt(w(vt(x, m_1 m_2 y), w(vt(x, m_1 m_2 m_2 y), vt(x, m_2 m_2 m_2 y))), m_1 y)$$

where $pr(x)$ is defined so that $pr(w(sx, y)) = w(x, y)$.

$vu(x, y)$ is the value of the function denoted by the unary functor number y for the argument x; $vb(x, y)$ is the value of the function denoted by the binary functor number y for the arguments $m_1 x$ and $m_2 x$; $vt(x, y)$ is the value of the function denoted by the ternary functor number y for the arguments $m_1 x$, $m_1 m_2 x$ and $m_2 m_2 x$.

3.3 We define now the functions which give the valuation of the terms and equations of T. From this point all definitions can be shown to be given by composition and primitive recursion only.

$$vo(x, 4y) = n(x, y)$$
$$vo(x, 4y + 1) = vu(vo(x, m_2 y), m_1 y)$$
$$vo(x, 4y + 2) = vb(w(vo(x, m_1 m_2 y), vo(x, m_2 m_2 y)), m_1 y)$$
$$vo(x, 4y + 3) = vt(w(vo(x, m_1 m_2 y), w(vo(x, m_1 m_2 m_2 y), vo(x, m_2 m_2 m_2 y))), m_1 y)$$

where $n(x, y) = m_2 n'(x, y)$ and $n'(x, y)$ is given by $n'(x, 0) = w(x, 0)$ and $n'(x, sy) = m_1 n'(x, y)$.

$vo(x, y)$ is the value of the term number y for the valuation number x of its numerical variables, where the valuation number x means that the value $m_2 x$ is assigned to the variable a, the value $m_2 m_1 x$ to b, the value $m_2 m_1 m_1 x$ to c, the value $m_2 m_1 m_1 m_1 x$ to a_1 and so on. (Note that as $m_1 x + m_2 x \leq x$, this is a valid definition.)

$$vl(x, y) = a(vo(x, m_1 y), vo(x, m_2 y))$$

$vl(x, y)$ has the value 0 or 1 according as the equation number y for the valuation number x of its variables (in the sense above) is true or false. The correspondence between these definitions and the numbering of the objects of T, given in 3.1, should be carefully noted.

3.4 We now define some auxiliary functions which will lead to the definition of $th(y)$. Each of these functions is associated with an axiom schema and enumerates the results of applying this schema to every term or equation of the system.

In the next section we require the subsidiary function $ws(x, y, z)$ defined by

$$ws(x, y, 0) = w(m_1 x, y)$$
$$ws(x, y, sz) = w(ws(m_1 x, y, z), m_2 x).$$

Theorem 1. $$n(ws(u, x, y), z) = x \quad \text{if } sy = z$$
$$= n(u, z) \quad \text{if } sy \neq z$$

This is obvious when $z = 0$, and we apply induction on y when $z > 0$ using the result $n(m_1 x, y) = n(x, sy)$.

The function associated with substitution is defined by,

$$st(x, y, 4z) = x \qquad \text{if } sy = z$$
$$st(x, y, 4z) = 4z \qquad \text{if } sy \neq z$$
$$st(x, y, 4z + 1) = 4w(m_1 z, st(x, y, m_2 z)) + 1$$
$$st(x, y, 4z + 2) = 4w(m_1 z, w(st(x, y, m_1 m_2 z), st(x, y, m_2 m_2 z))) + 2$$
$$st(x, y, 4z + 3)$$
$$= 4w(m_1 z, w(st(x, y, m_1 m_2 z), w(st(x, y, m_1 m_2 m_2 z), st(x, y, m_2 m_2 m_2 z)))) + 3$$

$st(x, y, z)$ is the number of the term which results from the substitution of term number x for the $(y + 1)$th variable in term number z.

Theorem 2. $$vo(u, st(x, y, z)) = vo(ws(u, vo(u, x), y), z)$$

Let $P(z)$ stand for this equation, $P(4z)$ follows immediately from the definition of $vo(x, y)$ and theorem 1. From the definitions we have

$$vo(u, st(x, y, 4z + 1)) = vu(vo(u, st(x, y, m_2 z)), m_1 z)$$

and

$$vo(ws(u, vo(u, x), y), 4z + 1) = vu(vo(ws(u, vo(u, x), y), m_2 z), m_1 z)$$

hence

$$P(m_2 z) \to P(4z + 1).$$

By exactly similar argument we derive

$$P(m_1 m_2 z) \,\&\, P(m_2 m_2 z) \quad \to P(4z + 2)$$

and

$$P(m_1 m_2 z) \,\&\, P(m_1 m_2 m_2 z) \,\&\, P(m_2 m_2 m_2 z) \to P(4z + 3)$$

and theorem 2 now follows.

$se(x, y, z)$ is given by the equation,

$$se(x, y, z) = w(st(x, y, m_1 z), st(x, y, m_2 z))$$

and is the number of the equation which results from the substitution of term number x for the $(y + 1)$th variable in equation number z.

Theorem 3. $\quad vl(u, se(x, y, z)) = vl(ws(u, vo(u, x), y), z)$

This is an immediate consequence of theorem 2.
We shall abbreviate $st(x, 0, y)$ to $st(x, y)$ and $se(x, 0, y)$ to $se(x, y)$.

3.5 We define $ax(y)$ as the number of the $(y + 1)$th axiom of T, as listed in section 2, if $y \leqq 10$ and 0 otherwise.

Theorem 4. $\quad vl(x, ax(r(y, 11))) = 0$

The formal definition of $ax(y)$ and the proof of theorem 4 will be omitted; the detailed work is simple but very long.

We define $tr(x, y)$ by the conditions,

$$tr(x, y) = w(m_2 x, m_2 y) \quad \text{if } m_1 x = m_1 y$$
$$tr(x, y) = x \quad \text{otherwise.}$$

$tr(x, y)$ is the number of the equation inferred from equations numbered x and y by axiom schema T.

Theorem 5. $\quad vl(x, y) = 0 \;\&\; vl(x, z) = 0 \rightarrow vl(x, tr(y, z)) = 0$

By considering the cases $m_1 y = m_1 z$ and $m_1 y \neq m_1 z$ this is readily proved.

$us(x, y)$ is defined by the conditions,

$$us(x, y) = w(st(m_1 x, y), st(m_2 x, y)) \quad \text{if } r(y, 4) = 3 \text{ and } m_2 q(y, 4) = w(4, w(8, 12))$$
$$us(x, y) = x \quad \text{otherwise.}$$

$us(x, y)$ is the number of the equation inferred by axiom schema U from equation number x and term number y.

Theorem 6. $\quad vl(x, y) = 0 \rightarrow vl(x, us(y, z)) = 0$

If $r(z, 4) = 3 \;\&\; m_2 q(z, 4) = w(4, w(8, 12)) \;\&\; vl(x, y) = 0$, then

$$vl(x, us(y, z)) = a(vo(ws(x, vo(x, m_1 y), 0), z), vo(ws(x, vo(x, m_2 y), 0), z)) = 0$$

and the theorem follows.

3.6 Now we define the function associated with E:

$$es(y) = w(m_2 y, st(0, m_2 y)) \quad \text{if } m_1 y = st(101, m_2 y),\ r(m_2 y, 4) = 3 \text{ and } m_2 q(m_2 y, 4) = w(4, w(8, 12))$$
$$es(y) = y \quad \text{otherwise.}$$

$es(y)$ is the number of the equation inferred from the equation number y by axiom schema E.

Theorem 7. $\sum_{t}^{m_2 x} [vl(w(m_1 x, t), y)] = 0 \rightarrow vl(x, es(y)) = 0.$

We require two subsidiary results, which are easily proved.

(a) $$vo(w(x, y), 101) = y + 1$$

(b) $$\sum_{t}^{x} [a(f(t+1), f(t))] = 0 \rightarrow a(f(x), f(0)) = 0$$

where $f(t)$ is any function. We let A stand for the conjunction of the premises in the first line of the definition of $es(y)$.

Now we have, using the definitions of $vo(x, y)$ and $ws(x, y, z)$

$$\begin{aligned} m_1 y = st(101, m_2 y) \rightarrow vl(w(m_1 x, t), y) & \\ = a(vo(w(m_1 x, t), st(101, m_2 y)), vo(w(m_1 x, t), m_2 y)) & \\ = a(vo(w(m_1 x, t+1), m_2 y), vo(w(m_1 x, t), m_2 y)) & \qquad \text{by (a)} \end{aligned}$$

therefore, by (b)

$$A \mathbin{\&} \sum_{t}^{m_2 x} [vl(w(m_1 x, t), y)] = 0 \rightarrow a(vo(x, m_2 y), vo(w(m_1 x, 0), m_2 y)) = 0.$$

We have, applying theorem 2

$$A \rightarrow vl(x, es(y)) = a(vo(x, m_2 y), vo(w(m_1 x, vo(x, 0)), m_2 y))$$

and hence, as the result is clearly true if A is false, the theorem follows.

Before we introduce the function associated with F we define $ft(y)$ as

$$ft(y) = 4w(m_1 q(y, 4), w(4, m_2 m_2 q(y, 4))) + 2.$$

If y is the number of a term of the form $f(A, B)$, then $ft(y)$ is the number of the term $f(a, B)$ where a is the first numerical variable.

Theorem 8.

$$r(y, 4) = 2 \mathbin{\&} m_1 m_2 q(y, 4) = z \mathbin{\&} m_2 m_2 q(y, 4) = 8 \rightarrow st(z, ft(y)) = y$$

Applying the definitions of $ft(y)$ and $st(x, y)$ this result is easily proved.

$fs(x, y, z)$ is defined by the conditions,

$$\begin{aligned} fs(x, y, z) = w(ft(m_1 y), ft(m_1 z)) & \\ \text{if } m_1 x = st(0, ft(m_1 y)), m_2 x = st(0, ft(m_1 z)), & \\ m_1 y = 4w(f, w(101, 8)) + 2, m_1 z = 4w(g, w(101, 8)) + 2, & \\ m_2 y = 4w(2w(p, w(m_1 q(ft(m_1 y), 4), 0)) + 2, w(4, 8)) + 2 \text{ and} & \\ m_2 z = 4w(2w(p, w(m_1 q(ft(m_1 z), 4), 0)) + 2, w(4, 8)) + 2 & \\ fs(x, y, z) = x \qquad \text{otherwise;} & \end{aligned}$$

$fs(x, y, z)$ is the number of the equation inferred from equations numbered x, y and z by axiom schema F.

Theorem 9.

$$vl(u, x) = 0 \ \& \sum_t^{m_1 u} [vl(w(m_1 u, t), y)] = 0 \ \& \sum_t^{m_1 u} [vl(w(m_1 u, t), z)] = 0 \rightarrow vl(u, fs(x, y, z)) = 0$$

As the derivation of this result is long and similar to that of theorem 7 only a brief outline is given. We require the subsidiary result (c):

$$a(f(0), g(0)) = 0 \ \& \sum_t^{x} [a(f(t+1), h(f(t)))] = 0 \ \& \sum_t^{x} [a(g(t+1), h(g(t)))] = 0 \rightarrow a(f(x), g(x)) = 0$$

We let A stand for the conjunction of the premises in the first line of the definition of $fs(x, y, z)$. We then derive the following, where $f(t)$ stands for $vo(w(m_1 u, t), ft(m_1 y))$, $g(t)$ stands for $vo(w(m_1 u, t), ft(m_1 z))$ and $h(t)$ stands for $vt(w(t, w(m_2 m_1 u, 0)), p)$, by theorem 8, result (a) in the proof of theorem 7 and the definitions of the valuation functions,

$$A \ \& \ vl(w(m_1 u, t), y) = 0 \rightarrow a(f(t+1), h(f(t))) = 0 \qquad \text{i.}$$

$$A \ \& \ vl(w(m_1 u, t), z) = 0 \rightarrow a(g(t+1), h(g(t))) = 0 \qquad \text{ii.}$$

$$A \ \& \ vl(u, x) = 0 \rightarrow a(f(0), g(0)) = 0 \qquad \text{iii.}$$

Applying result (c) to (i), (ii) and (iii) theorem 9 follows, as it is clearly true if A is false.

3.7 We are now in a position to define $th(y)$

$$th(6y) = ax(r(y, 11))$$

$$th(6y+1) = se(m_1 y, m_1 m_2 y, th(m_2 m_2 y))$$

$$th(6y+2) = us(m_1 y, th(m_2 y))$$

$$th(6y+3) = tr(th(m_1 y), th(m_2 y))$$

$$th(6y+4) = es(th(y))$$

$$th(6y+5) = fs(th(m_1 y), th(m_1 m_2 y), th(m_2 m_2 y))$$

$th(y)$ is the number of the yth theorem in a certain enumeration of the theorems of the system T.

Theorem 10. $vl(x, th(y)) = 0.$

Let $V(x, y) = \sum_t^{6y+5} [vl(x, th(t))]$. $V(x, 0)$ is easily shown to be true. Applying theorems 4, 3, 6, 5, 7 and 9 we have,

$$\sum_t^{x + ws(x, vo(x, m_1(y+1)), m_1 m_2(y+1))} [V(t, y)] = 0 \rightarrow V(x, sy) = 0$$

thus $V(x, y) = 0$ follows (by the induction schema given in [7]). Theorem 10 is now given immediately.

3.8 To prove the freedom from contradiction of the system T in T^*, we require one further function $e(y)$: If y is the number of the equation $A = B$, then $e(y)$ is the number of the equation $A \dashv\vdash B$.

Theorem 11. $$vl(x, y) = 0 \rightarrow vl(x, e(y)) = 1$$

The formal definition of the function $e(y)$ is apparent and, from this, theorem 11 is readily deduced.

Theorem 12. $$vl(x, e(th(z))) = 1$$

This is an immediate consequence of theorems 10 and 11.

Theorem 13. $$th(y) \dashv\vdash e(th(z))$$

This result, deduced directly from 10 and 12, expresses the fact that the number of a theorem can never be the number of a non-theorem, that is, as every number has a unique equation associated with it, system T is free from contradiction.

4. Gödel's Incompleteness Theorems in System T

In this section we shall prove that there is an equation of system T which is undecidable in T and, from this, that the equation

$$th(y) \dashv\vdash e(th(z))$$

is also undecidable in T. This second result will be proved by showing that the proof of the corresponding result for systems containing the first order predicate calculus with equality, given in [4 pp. 283—289], can be carried out, in a modified form, in T. We shall write $[A]$ for the number of the object A, as given in 3.1.

We begin by proving the following meta-theorems.

Theorem 14. *If $A \vdash B$ in system T, then*

$$th(y) \dashv\vdash [B] \vdash th(y) \dashv\vdash [A]$$

in system T, where A and B are equations.

Suppose, first, that B is inferred from A by a substitution of the term number x for the $(z + 1)$th variable in A, then

$$[B] = se(x, z, [A])$$

and

$$th(y) \dashv\vdash [B] \vdash th(6u + 1) \dashv\vdash se(x, z, [A]),$$

now let $u = w(x, w(z, y))$, and it follows by the definition of $th(y)$ in 3.7 that

$$th(y) \dashv\vdash [B] \vdash se(x, z, th(y)) \dashv\vdash se(x, z, [A])$$
$$\vdash th(y) \dashv\vdash [A].$$

If B is inferred from A by axiom schema T, U, E or F exactly similar arguments will apply, thus, by repeated applications of these results, theorem 14 holds for any sequence of inferences from A to B.

Theorem 15. *If* $A = B$ *then*

$$th(y) \doteqdot [A = 0] \vdash th(y) \doteqdot [B = 0],$$

where A *and* B *are terms.*

This is an immediate consequence of theorem 14.

Theorem 16. $th(y) \doteqdot [f(0) = 0]$ & $th(y) \doteqdot [f(sx) = 0] \vdash f(x) \doteqdot 0$,

where $f(x)$ *is a 1, 2 or 3-function.*

We show first that the theorem is true for the initial functions. Clearly it holds for $s(x)$ We consider $i(x)$; by the definition of $th(y)$ we have

$$th(2347) = [i(0) = 0]$$

but

$$th(y) \doteqdot [i(0) = 0] \vdash th(i(x) + 2347) \doteqdot [i(0) = 0]$$
$$\vdash i(x) \doteqdot 0.$$

Very similar arguments apply for o, j and k.

Consider now the case of the function $Rfgh(x, u, z)$, defined by axioms 5 and 6 see section 2), we shall show that, if the result holds for f, g and h, it also holds for $Rfgh$. We have by theorem 15,

$$th(y) \doteqdot [Rfgh(0, u, z) = 0] \vdash th(y) \doteqdot [f(u, z) = 0]$$

and

$$th(y) \doteqdot [Rfgh(sx, u, z) = 0] \vdash th(y) \doteqdot [g(x, y) = 0]$$

from these and the hypotheses follow

$$f(u, z) \doteqdot 0 \quad \text{and} \quad g(x, y) \doteqdot 0$$

and thus

$$Rfgh(x, u, z) \doteqdot 0.$$

By similar reasoning the result can be proved for functions defined by R', C, C' or K and hence, the theorem holds for any particular function of the system T.

We define $nu(x)$ by,

$$nu(0) = 0 \quad \text{and} \quad nu(sx) = 4w(2, nu(x)) + 1$$

$nu(x)$ is the number of the numeral denoting x, as given in 3.1.

Theorem 17. $[se(nu(N), N) \doteqdot th(sy)] = se(nu(N), N)$

where $N = [se(nu(a), a) \doteqdot th(sy)]$

We have, by results in 3.4,

$$[se(nu(N), N)] = st(nu(N), [se(nu(a), a)])$$

and

$$[th(sy)] = st(nu(N), [th(sy)]),$$

and from the definitions of the functions involved,

$$[se(nu(N), N) \doteqdot th(sy)] = e(w([se(nu(N), N)], [th(sy)]))$$

and

$$se(nu(N), N) = e(w(st(nu(N), [se(nu(a), a)]), st(nu(N), [th(sy)]))),$$

theorem 17 is readily deduced from these equations. It is easily seen, from this result, that

$$se(nu(N), N) \dashv\vDash th(sy)$$

is undecidable in system T, if T is consistent. This is GÖDELS first incompleteness result for T, the second is given by the following theorem.

Theorem 18. *$th(x) \dashv\vDash e(th(z))$ & $th(y) = se(nu(N), N) \vdash th(y) \dashv\vDash se(nu(N), N)$, where N is defined in the preceding result.*

This is proved by applying theorem 16 to $a(se(nu(N), N), th(y)) = 0$.

(i) As $se(nu(N), N)$ is not the number of the first axiom, there is an X such that

$$th(X) = [se(nu(N), N) \dashv\vDash th(0)]$$

so by the first premise and the definition of $e(y)$,

$$th(x) \dashv\vDash e(th(z)) \vdash e(th(z)) \dashv\vDash [se(nu(N), N) \dashv\vDash th(0)]$$
$$\vdash th(z) \dashv\vDash [se(nu(N), N) = th(0)].$$

(ii) From the two premises we have

$$e(th(z)) \dashv\vDash se(nu(N), N)$$

and by theorem 17,

$$e(th(z)) \dashv\vDash [se(nu(N), N) \dashv\vDash th(sy)]$$

and

$$th(z) \dashv\vDash [se(nu(N), N) = th(sy)].$$

Thus by (i), (ii), the definition of $a(x, y)$ and theorem 16 the result is given. If

$$th(y) \dashv\vDash e(th(z))$$

is a theorem of T, then by the previous result so is

$$se(nu(N), N) \dashv\vDash th(sy)$$

but by 17 this implies that T is inconsistent, thus if T is free from contradiction

$$th(y) \dashv\vDash e(th(z))$$

is undecidable in T

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(Eingegangen am 24. März 1961)